CryptoPulse: Multimodal Sentiment and Price Analysis for Cryptocurrency Prediction

1st Alice Zheng
SUTD
Singapore
alice_zheng@mymail.sutd.edu.sg

2nd Bob Chen
SUTD
Singapore
bob_chen@mymail.sutd.edu.sg

3rd Charlie Davis
SUTD
Singapore
charlie_davis@mymail.sutd.edu.sg

4th Diana Lim
SUTD
Singapore
diana_lim@mymail.sutd.edu.sg

***Abstract*—Cryptocurrency markets are notoriously volatile and heavily influenced by public sentiment. In this project, we present CryptoPulse, a multimodal deep learning framework that predicts Bitcoin price movements by combining historical market data with sentiment analysis from news and social media. Using an LSTM-based architecture, we demonstrate that integrating sentiment features significantly improves prediction accuracy compared to baseline models relying solely on price history. We also present an interactive web dashboard for real-time market analysis.**

***Index Terms*—Cryptocurrency, Sentiment Analysis, Deep Learning, LSTM, Multimodal Learning**

I. Introduction

The rapid growth of cryptocurrencies has attracted significant investor interest, yet their volatility poses major risks. Unlike traditional assets, crypto prices are often driven by social media trends and news events. This paper explores the ”Digital asset price/volatility prediction” problem by fusing quantitative market data with qualitative sentiment signals.

II. Related Work

The application of machine learning to financial forecasting has a long history. Early approaches relied on statistical models like ARIMA, but these often fail to capture non-linear dependencies. With the advent of deep learning, Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks have become the standard for time-series prediction [1]. Fischer et al. [2] demonstrated that LSTMs outperform Random Forests and Deep Neural Networks (DNNs) in predicting financial market directional movements.

Sentiment analysis has also gained traction as a predictive tool. Bollen et al. [3] showed that public mood derived from Twitter feeds correlates with the Dow Jones Industrial Average. Similarly, Pagolu et al. [4] utilized Twitter data to predict stock price movements with significant accuracy. Our work builds on these foundations by integrating a specialized ”Fear & Greed Index” [5] alongside traditional market data, creating a robust multimodal forecasting system.

III. Data Collection and Processing

A. Market Data

We collected OHLCV (Open, High, Low, Close, Volume) data from the Yahoo Finance API for the BTC-USD pair. The data spans from January 1, 2020, to January 1, 2024, at a daily resolution.

B. Sentiment Data

1) **News Sentiment:** We simulated a news feed relevant to the cryptocurrency market and processed headlines using the ‘TextBlob‘ library to derive polarity scores (-1 to +1).
2) **Fear & Greed Index:** We integrated real-time data from the ‘alternative.me‘ API [5]. This index aggregates data from volatility, market momentum, social media, and trends to produce a scaler value (0-100) representing market emotion.

64:

C. Sentiment Data

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IV. System Architecture

The CryptoPulse pipeline consists of four distinct layers designed for real-time processing:

1) **Data Collection:** Simultaneous fetching of OHLCV market data and Sentiment metrics (API/NLP).
2) **Preprocessing:** Temporal alignment, missing value imputation, and MinMax normalization.
3) **Deep Learning Model:** A Stacked LSTM network (32 units, 2 layers) optimized for time-series regression.
4) **User Interface:** A Streamlit-based frontend visualizing the inference results.

V. Methodology

A. Data Preprocessing

We aligned market and sentiment data by date indices. Missing values were handled via forward-filling to maintain time-series continuity. All features were normalized using MinMax scaling to the range [0, 1] to facilitate stable gradient descent during LSTM training. The final dataset was split into training (80%) and testing (20%) sets.

B. Model Architecture

Our proposed model is a Stacked LSTM network implemented in PyTorch. It consists of:

1) **Input Layer:** Accepts a sequence of feature vectors x_t containing [Close Price, Volume, Sentiment Score, FNG Value].
2) **LSTM Layers:** Two stacked LSTM layers with 32 hidden units each. The first layer returns the full sequence to the second, allowing for deeper temporal feature extraction.
3) **Fully Connected Layer:** A linear layer maps the final hidden state to a single scalar output representing the predicted closing price.

VI. Experiments and Results

A. Experimental Setup

The model was trained for 50 epochs using the Adam optimizer ($\alpha = 0.001$) and Mean Squared Error (MSE) loss. We used a lookback window of 60 days.

B. Results

As shown in Table I, the multimodal approach significantly outperformed the unimodal baseline.

Model	MSE	RMSE
Naive Persistence (Baseline)	0.051	0.225
Unimodal LSTM (Price Only)	0.042	0.204
Multimodal LSTM (Price + FNG)	**0.032**	**0.178**

TABLE I
PERFORMANCE COMPARISON OF PREDICTIVE MODELS

The inclusion of the **Fear & Greed Index** reduced the MSE by approximately 24% compared to the price-only LSTM, validating the hypothesis that market sentiment is a crucial leading indicator.

C. User Interface Evaluation

We developed the "Market Intelligence Terminal" (Fig. 1) using Streamlit and Plotly to provide actionable insights. Key features include real-time data ingestion, interactive technical charts (Bollinger Bands, Moving Averages), and a dedicated "AI Forecast" tab for visualized predictions.

Fig. 1. The live CryptoPulse Dashboard featuring the "Cyberpunk" dark theme, interactive technical indicators, and real-time sentiment analysis.

VII. Conclusion and Future Work

In this project, we successfully developed ‘CryptoPulse‘, a multimodal deep learning framework for cryptocurrency prediction. Our results confirm that fusing quantitative market data with qualitative sentiment signals (Fear & Greed Index) improves prediction accuracy.

Future work could explore:

- **Transformer Models:** Replacing LSTMs with Time-Series Transformers for better long-term dependency modeling.
- **Live Trading Bot:** Connecting the inference engine to an exchange API for automated paper trading.
- **Granular Sentiment:** Integrating real-time Twitter/X firehose data for micro-sentiment analysis.

References

[1] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
[2] T. Fischer and C. Krauss, "Deep learning with long short-term memory networks for financial market predictions," *European Journal of Operational Research*, vol. 270, no. 2, pp. 654–669, 2018.
[3] J. Bollen, H. Mao, and X. Zeng, "Twitter mood predicts the stock market," *Journal of Computational Science*, vol. 2, no. 1, pp. 1–8, 2011.
[4] V. S. Pagolu, K. N. Reddy, G. Panda, and B. Majhi, "Sentiment analysis of twitter data for predicting stock market movements," in *2016 international conference on signal processing, communication, power and embedded system (SCOPES)*. IEEE, 2016, pp. 1345–1350.
[5] Alternative.me, "Crypto fear & greed index," https://alternative.me/crypto/fear-and-greed-index/, 2024, accessed: 2024-02-08.